Development and evaluation of gelatin starch based folic acid loaded orally disintegrating film

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BIOMEDICAL ENGINEERING

Background

Orally Disintegrating Film

Orally Disintegrating film (ODF) is a drug delivery platform which dissolve in the saliva within 180 s without the need of water.

- ❑ Patient convenient
- ❑ Avoids first pass metabolism of conventional oral dosages.



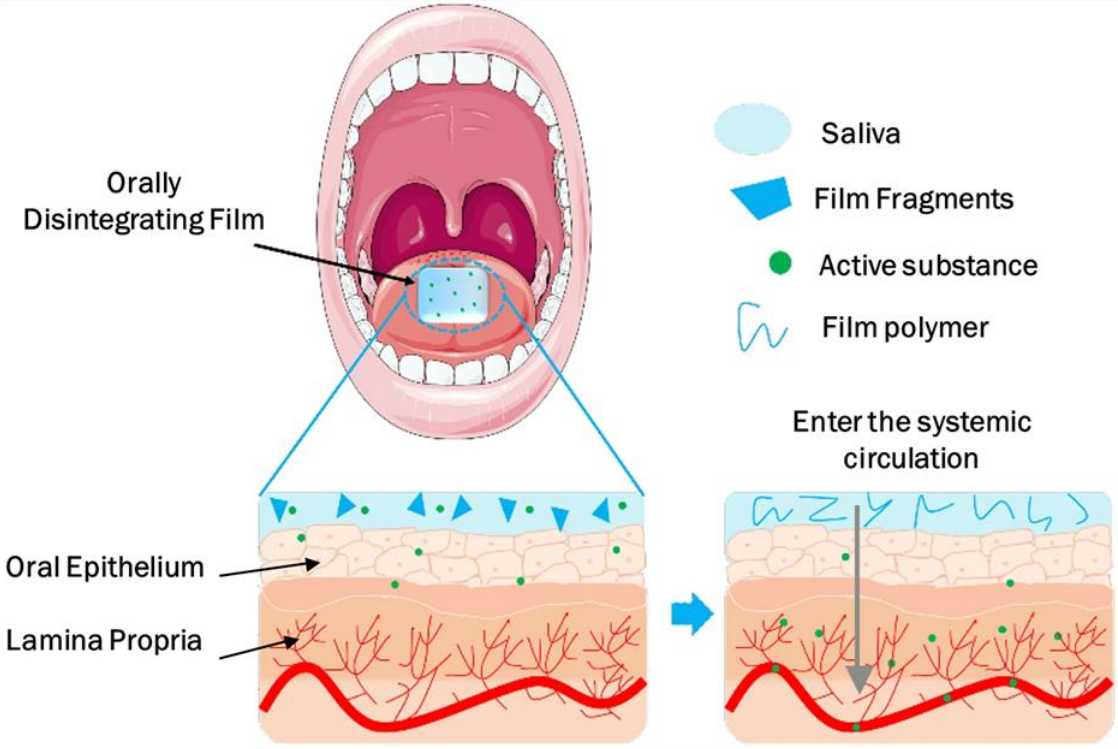
Background

Physiology of Oral Mucosa

Oral mucosa is highly vascularized with compare to GI tract and skin

Free from extreme pH, harsh enzymes and less reactive to drugs.

It has 4000 times more permeability despite of having a smaller surface area

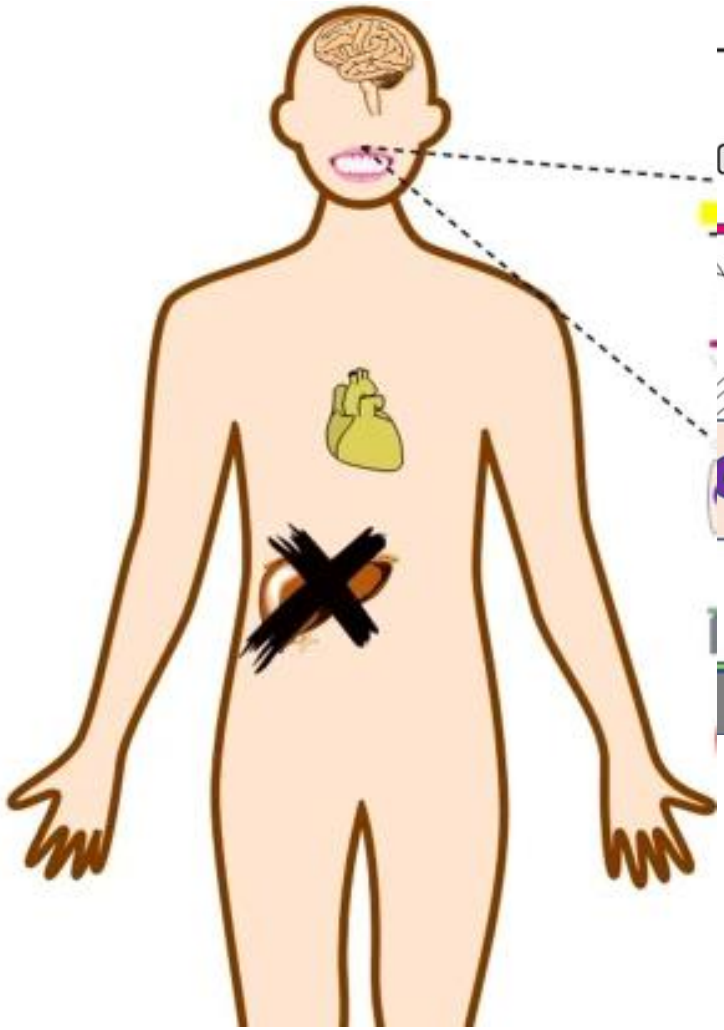


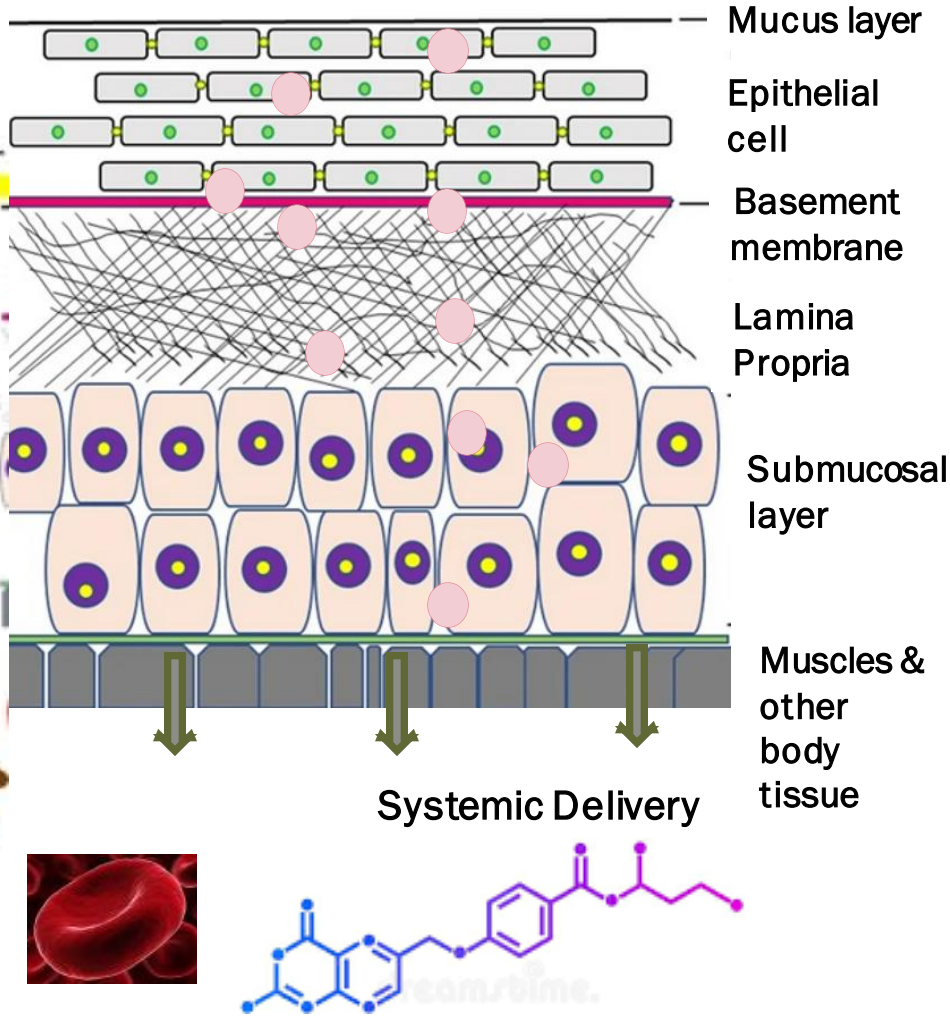
GI TRACT	30000 cm2
ORAL MUCOSA	200 cm2
SKIN	20000 cm2

1. Yuan, C., Sha, H. & Cui, B. Orally Disintegrating Film: A New Approach to Nutritional Supplementation. *Food Bioprocess Technol* **15**, 2629–2645 (2022).

Background

Circulation through Oral Mucosa





2. Suresh, J., Aravindaraj, N., Krishnaswami, V. *et al.* Oral film technology: a perspective on translational research and clinical studies. *Macromol. Res.* **31**, 917–932 (2023). https://doi.org/10.1007/s13233-023-00185-y

Problem Statement

Dysphagia is defined as impairment to swallowing, is a critical condition which affects elderly, especially with neurological disorders and children

Dysphagia can lead to serious complications such as choking, aspiration pneumonia, malnutrition and dehydration.

This study hypothesizes the development of an oral thin film using gelatin and modified sweet potato starch as the film matrix for improving the convenience of drug administration.

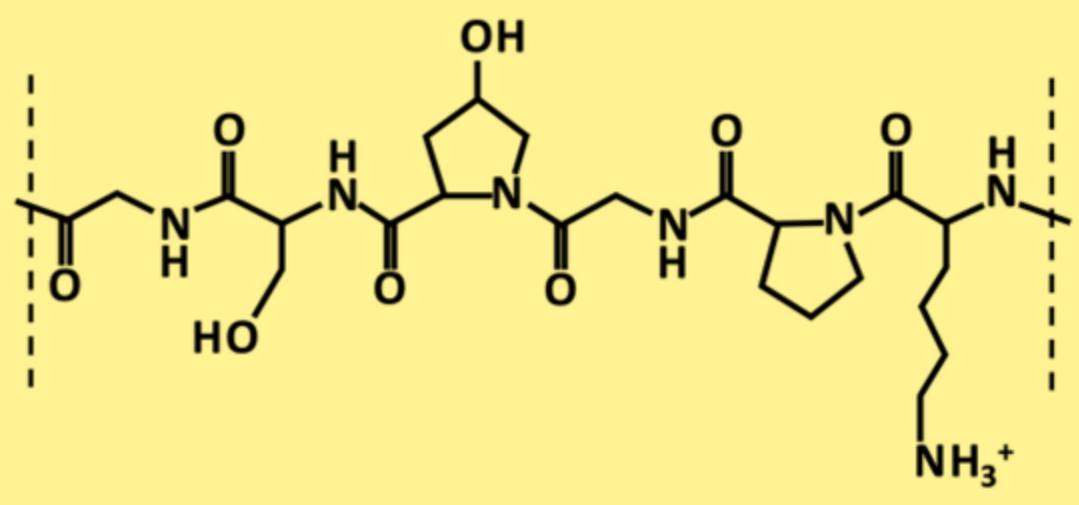
Objectives

- Extract starch from native sweet potato and modify the starch in physical and chemical methods.
- Develop orally disintegrating film based on gelatin and synthesized starch.
- Incorporate folic acid into the orally disintegrating film.
- Characterize and functionally evaluate the folic acid loaded gelatin-starch based ODF

Literature Gap

Why Gelatin?

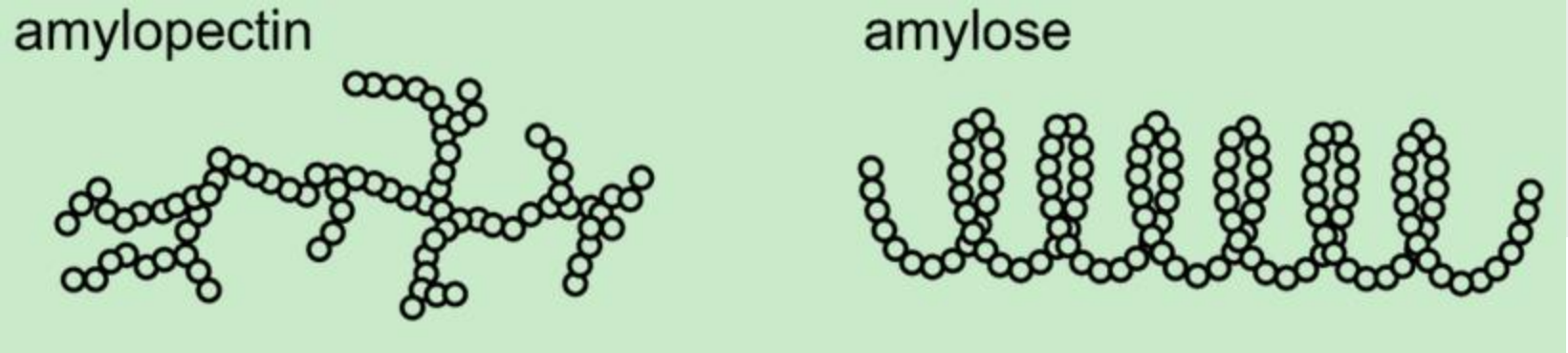
- Mucoadhesiveness
- Film forming ability
- Gel forming capacity
- Hydrophilicity



Literature Gap

Why Starch?

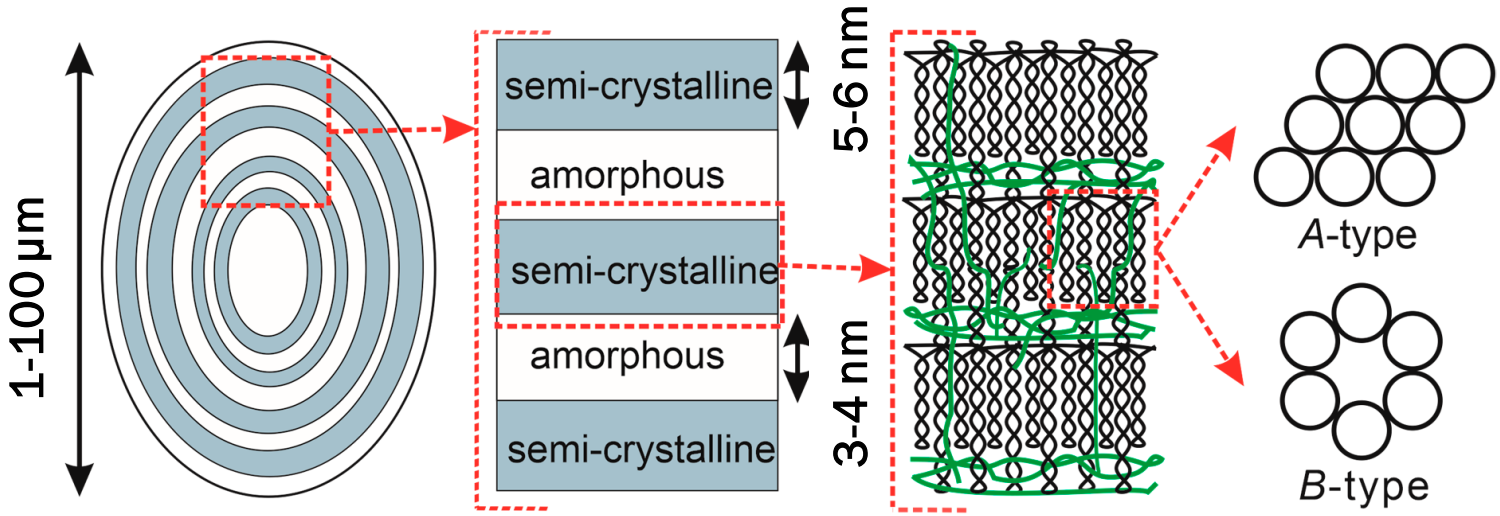
- Sweet potato (*Ipomoea batatas*) ranks as the seventh most significant food crop globally in terms of production.
- Works as disintregation enhancer.
- Film forming capacity
- There has been no study on how sweet potato starch modification influences ODF characteristics



Literature Gap

Why starch modification?

Starch structure from granule to glucoside



- Starch granules are consisted of alternating semi crystalline and amorphous region
- Crystalline region consists of amylopectin and amorphous region with interspersed amylose
- Higher crystallinity is linked with low reactivity, heat instability and low water solubility

Literature Gap

Why gelatin starch film ?

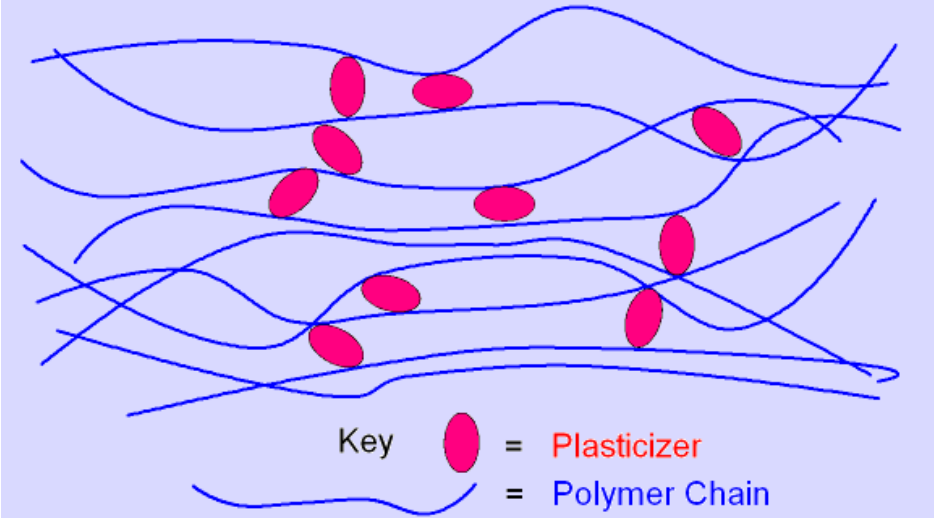
- Low cost and biodegradable
- The biocomposite has optimal mechanical properties and water vapor permeability.
- Extensively used in the food packaging industry



Literature Gap

Why plasticizer?

- Plasticizers are low molecular weight substances added to a polymer solution to promote its plasticity and flexibility.
- Plasticizer increase the free molecular space between polymer chains.
- Polyol (Glycerol, Sorbitol, PEG) used as plasticizer in gelatin starch based films.



Materials & Methods

Gelatin, Folic acid tablet, Sodium chloride (NaCl), potassium phosphate monobasic (KH_2PO4), disodium hydrogen phosphate (NaH_2PO4), ethanol(C_2H_5OH) and sodium hydroxide (NaOH)

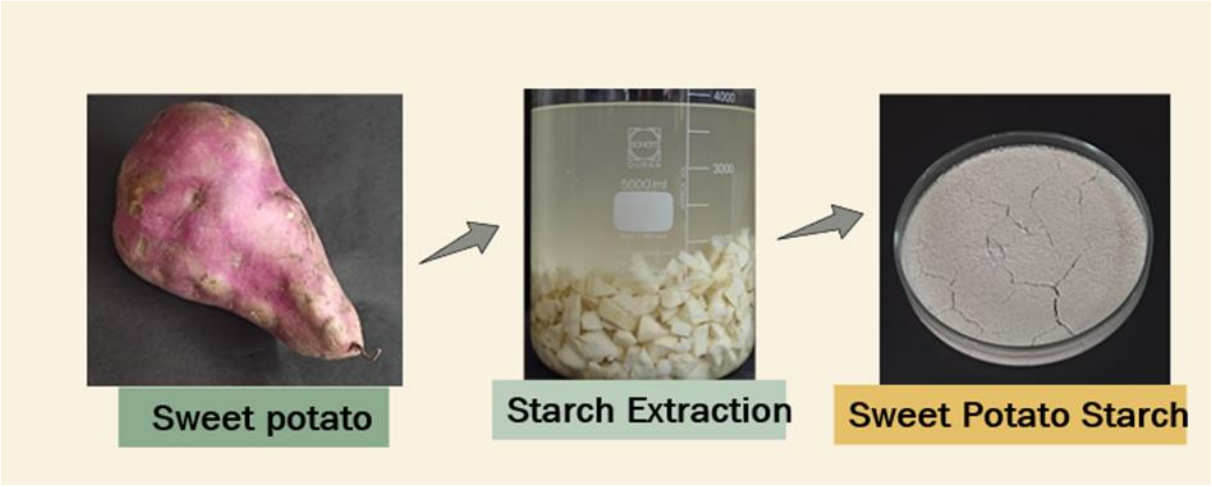
Simulated saliva (pH 6.85) was prepared by dissolving NaCl (8.00 g), disodium hydrogen phosphate NaH_2PO4 (2.38 g) and potassium phosphate monobasic (KH_2PO4) (0.19 g) in 1000 mL DI and HCl was used to adjust pH.

Experimental Overview

1.Extraction of starch

2. Physical modification of starch

3. Production of ODFs

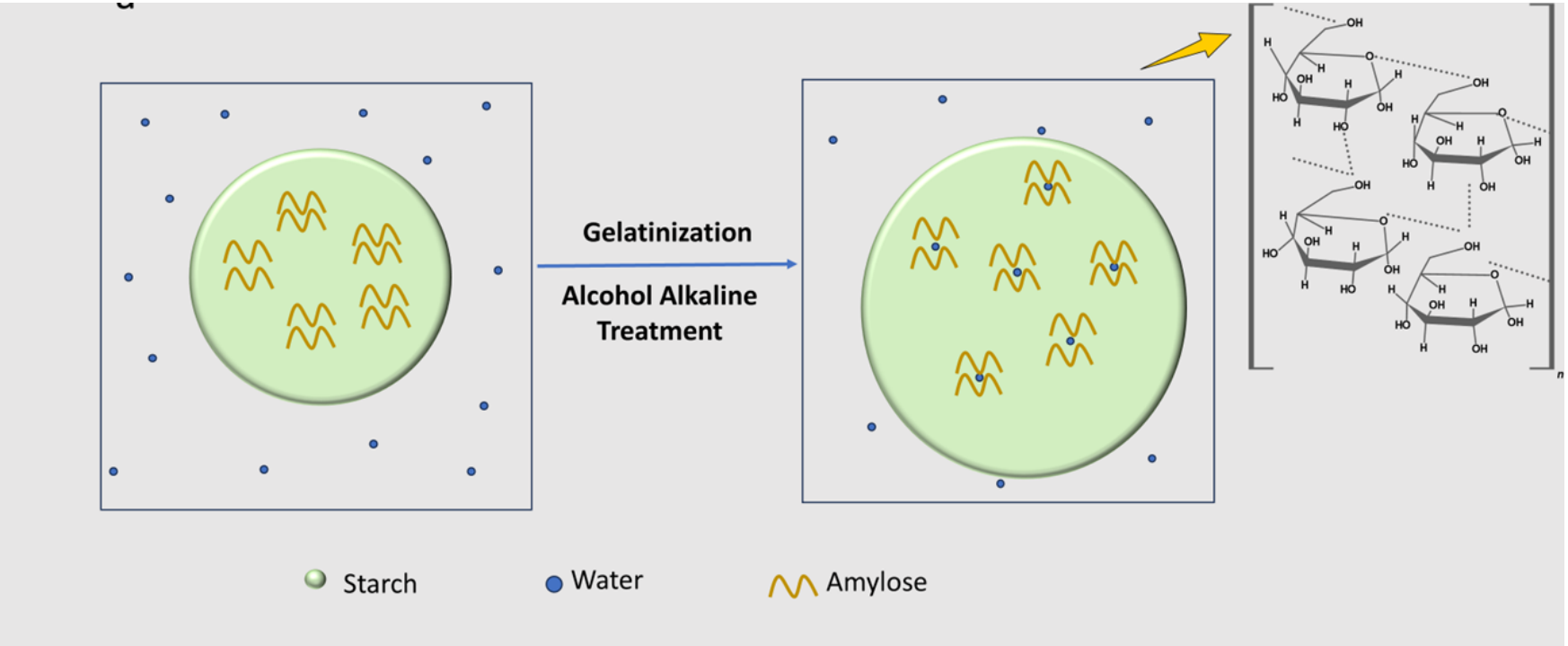


Part 1 of thesis

Physical Modification of Sweet Potato Starch by Pregelatinization & Alcohol Alkaline Method

Part 1 of thesis

Experimental Overview



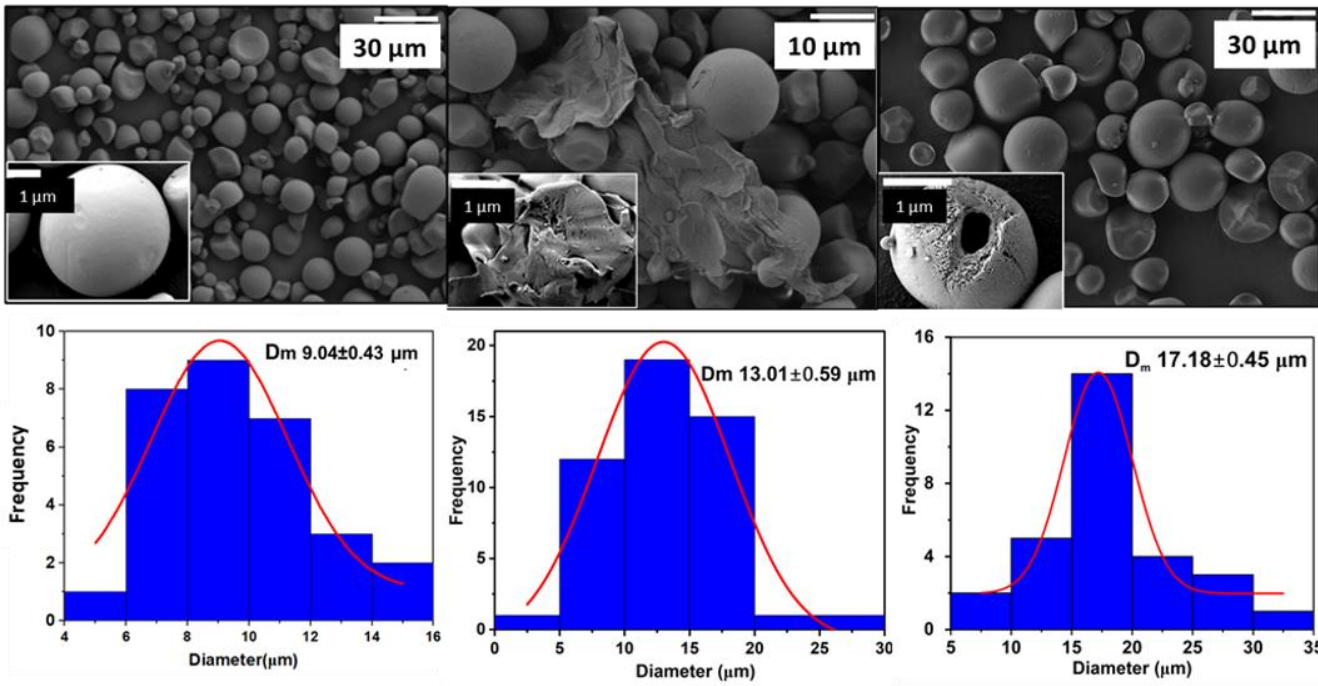
Part 1 of thesis

Results & Discussion

❖ Scanning Electron Microscopy

The native potato starch granules were oval shaped with smooth surface with mean diameter ranged 9.04 ± 0.043 µm

Modified sweet potato starch granules showed swollen structure and have larger particle size with rough, intended surface.



Results & Discussion

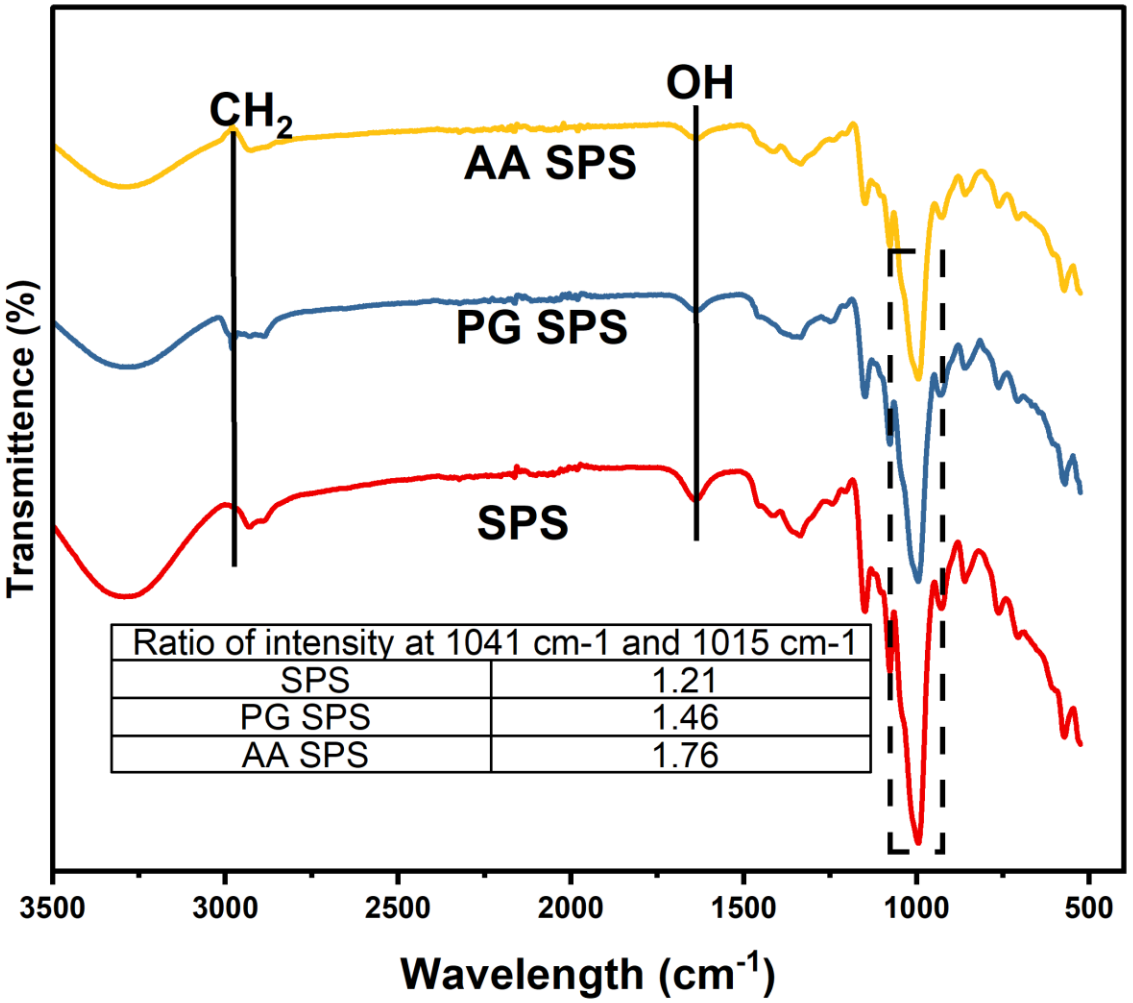
❖ FTIR

Starch modification **did not change the basic chemical structures** of sweet potato starch

The bands at **1015 and 1047** cm^{-1}, have been shown to be sensitive to change in starch crystalline

The intensity ratio of **1047/1015** cm^{-1} could be used to quantify the crystallinity

Ratio of 1047/1015 cm^{-1}: SPS> PG SPS> AA SPS



Ratio of intensity at 1041 cm-1 and 1015 cm-1	
SPS	1.21
PG SPS	1.46
AA SPS	1.76

Part 1 of thesis

Results & Discussion

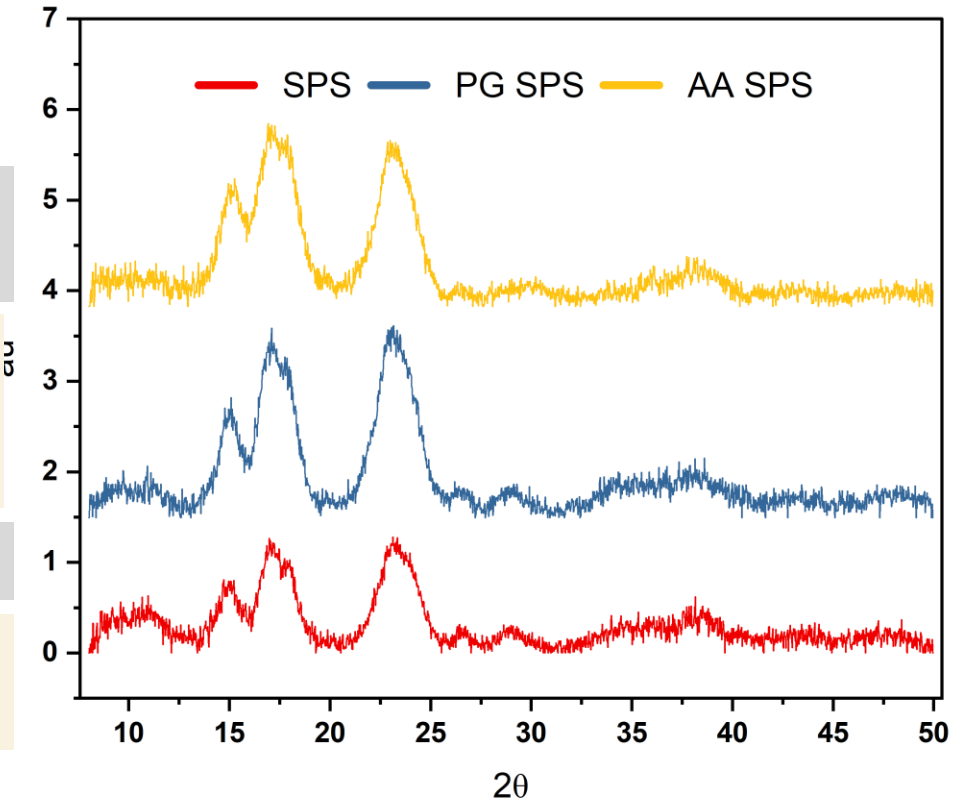
❖ XRD

The spectrum of all the starch showed peaks at 2θ=15°, 17° and 23°

Degree of crystallinity was calculated for SPS, PG SPS and AA SPS were 50.1, 22.5 and 17.8 respectively.

Sweet potato starch follow B pattern

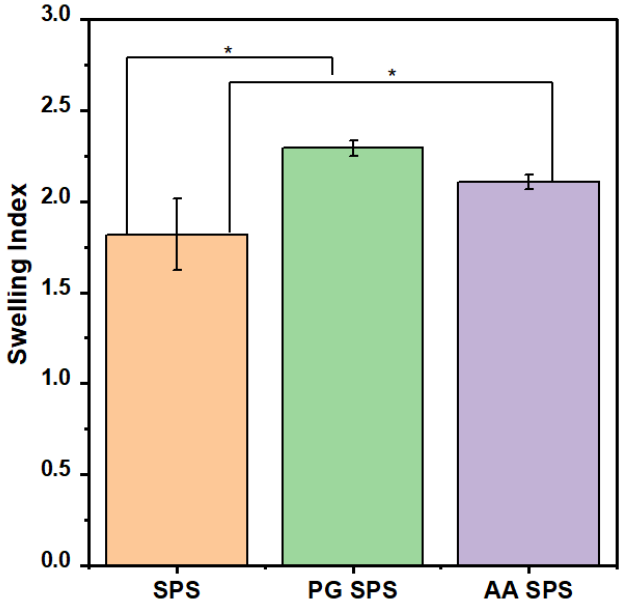
Modification process reduced the area crystalline region .

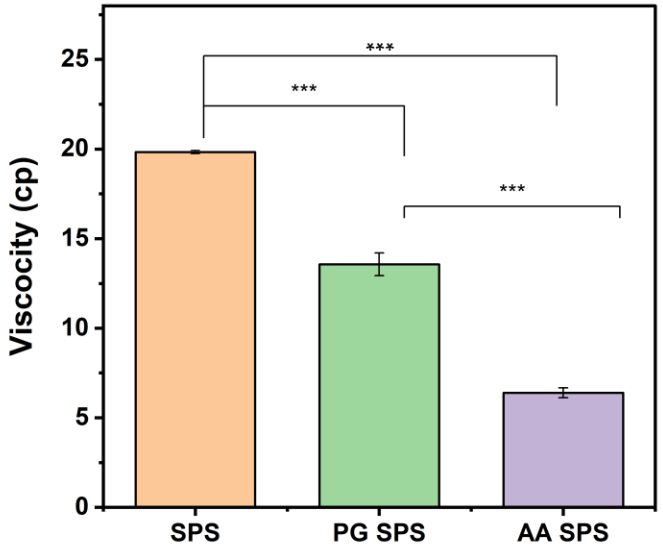


Part 1 of thesis

Results & Discussion

❖ Swelling & viscosity





After gelatinization and alcohol alkaline treatment, swelling index increased significantly

Disruption of crystalline regions> More prominent passage of water molecule into the inside of the starch granules

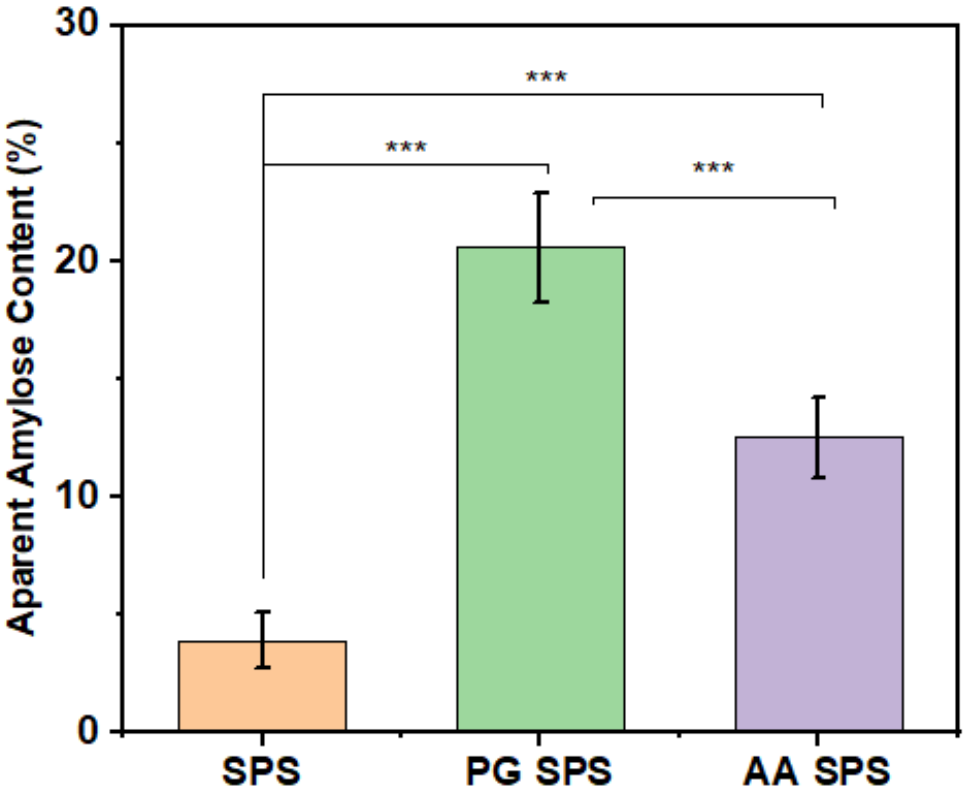
Gelatinization and alcohol alkaline treatment resulted in lower peak viscosity of the starch.

Results & Discussion

❖ Amylose Content

Amylose is an excellent film forming material with linear chains and stable network by hydrogen bonds

Physical treatments> cleavage of glycosidic bond of amylopectin > amylose content increased at the expense of loss of amylopectin.



Summary of Part 1 of thesis

Starch was extracted from native sweet potato

Starch was modified by two physical treatments: pregelatinization & alcohol alkaline treatment

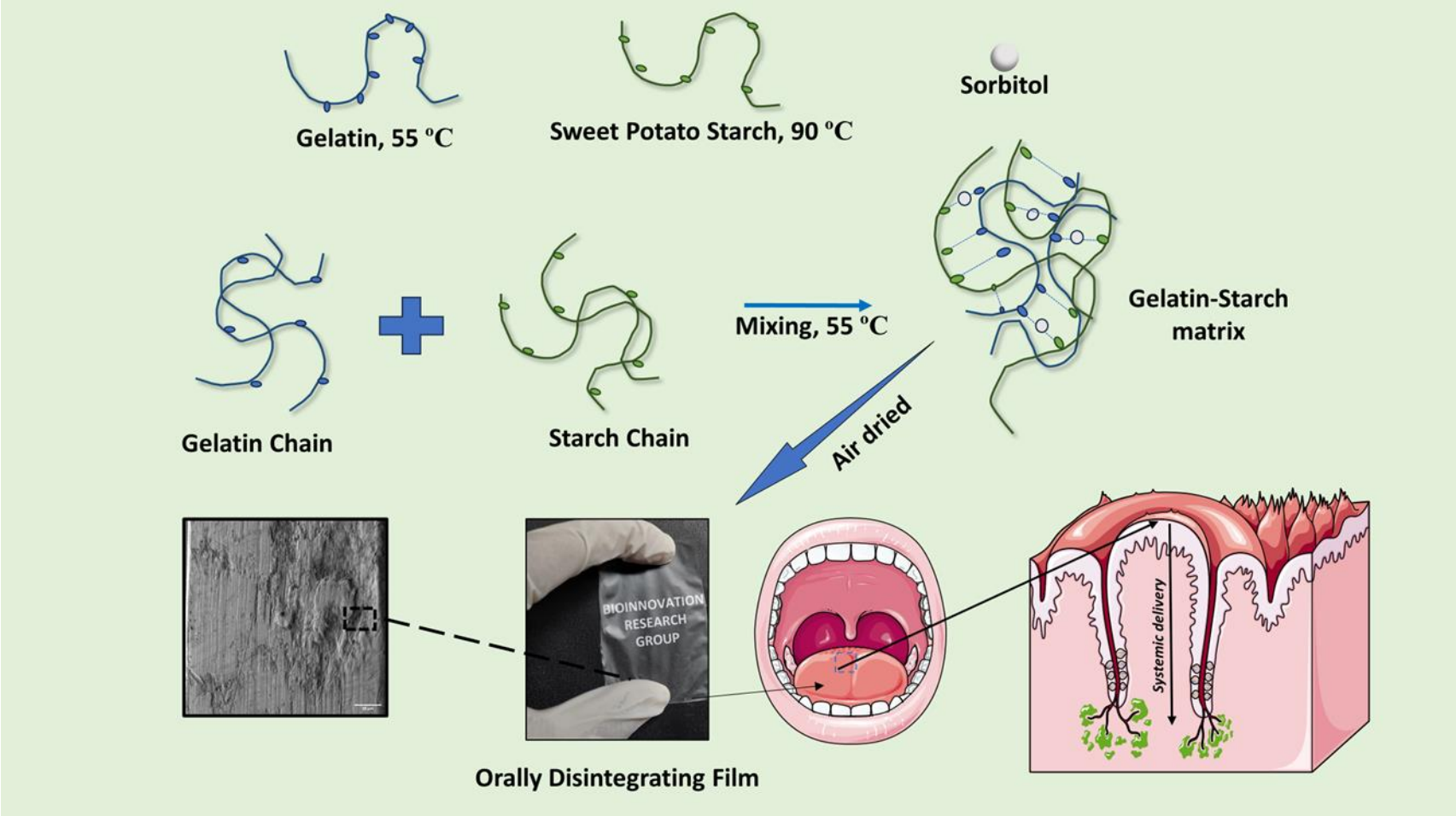
Physical modification of starch reduced the crystallinity and increased its solubility.

Part 2 of thesis

Development & evaluation of gelatin-modified starch based ODFs for raid release of folic acid

Part 2 of thesis

Experimental Overview



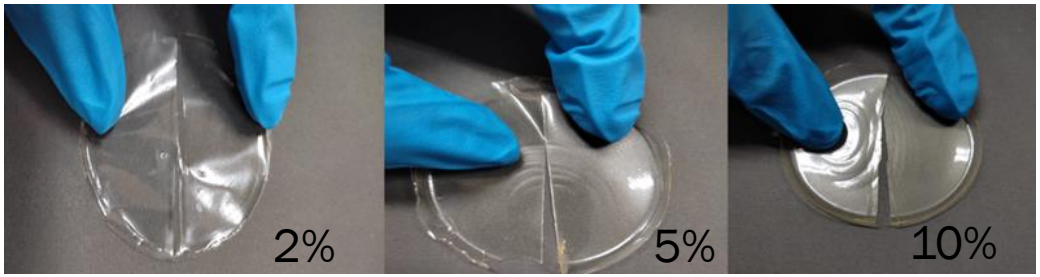
Part 2 of thesis

Results & Discussion

Optimizing process parameters

Total polymer concentration

Total concentration of polymer	Folding Endurance	Decision
10%	5	Not eligible
5%	20	Not eligible
2%	74	eligible



Part 2 of thesis

Results & Discussion

Optimizing process parameters

Total polymer concentration

Formulation	Disintegration time	Tensile Strength (MPa)	Casting volume	Decision
Porcine gelatin	6 hr	99.7		Not eligible
Fish gelatin	2s	3	3 mL	Not eligible
Bovine Gelatin	1.5 min	70		eligible

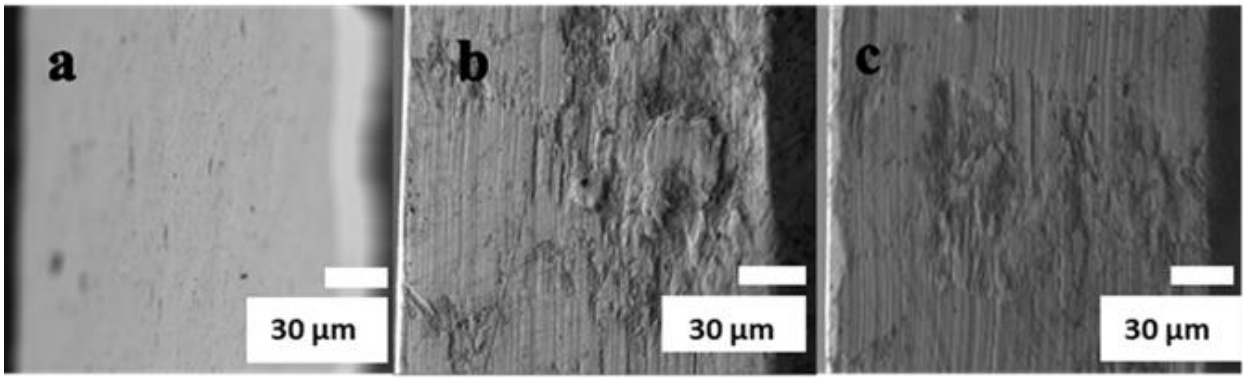
Part 2 of thesis

Results & Discussion

- SEM

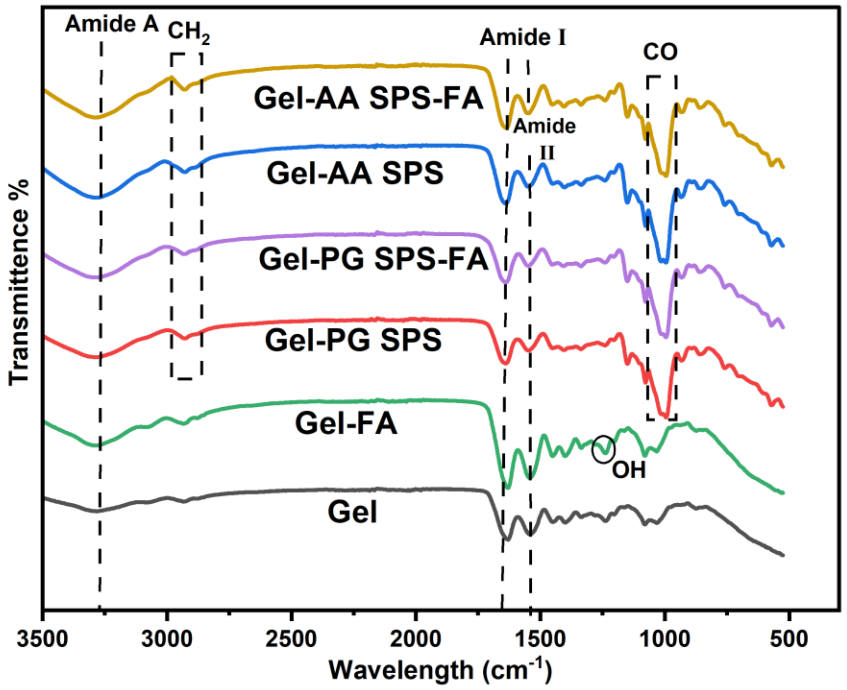
SEM of the cross sectional area of the ODFs shows heterogenous structure

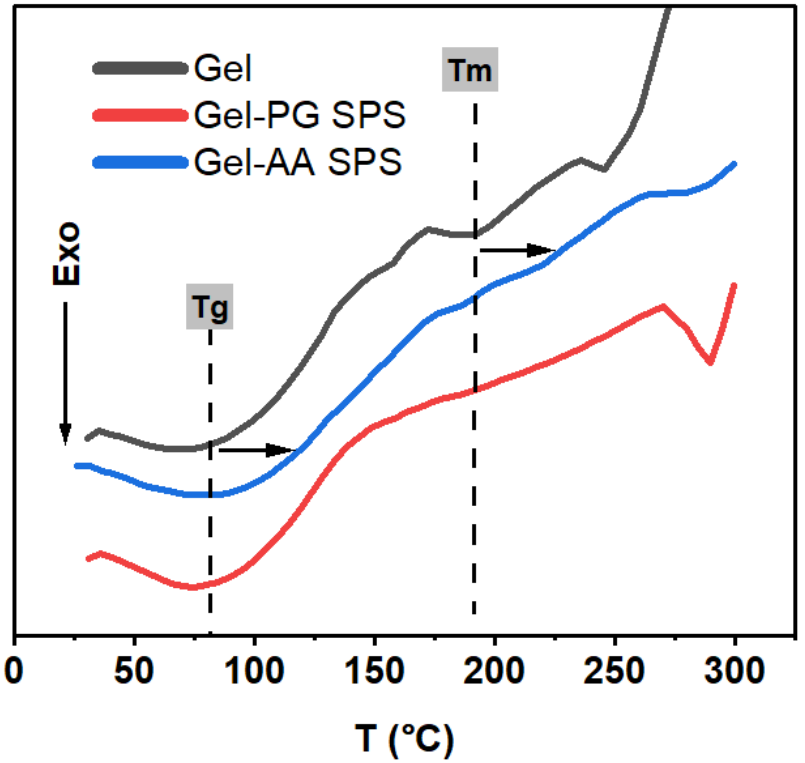
The introduction of starch into the films results in this heterogeneous surface due to the presence of starch aggregates



Results & Discussion

❖ FTIR & STA





ATR-FTIR showing characteristic peaks of starch, gelatin and folic acid

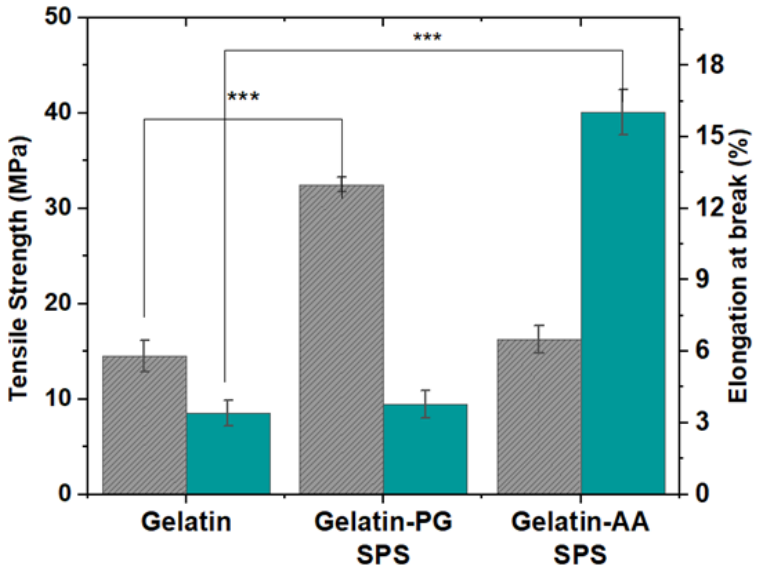
STA indicates two steps thermal degradation of ODFs.

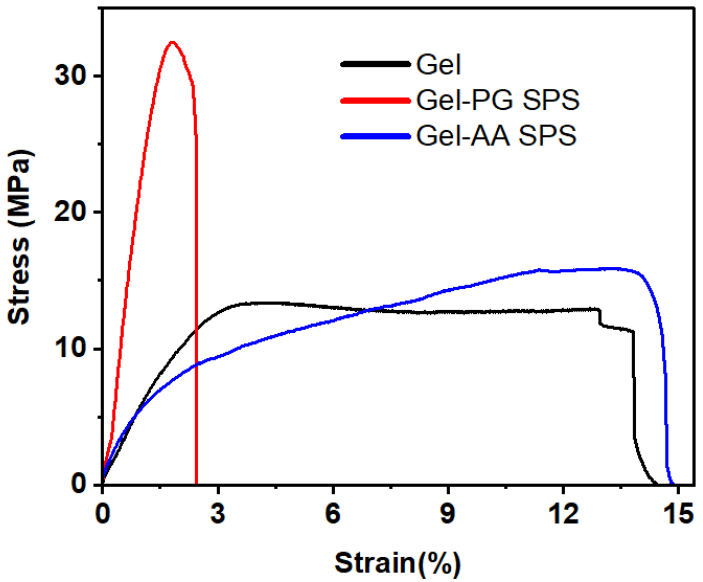
Only one Tg could indicate that the gelatin and starch in the ODFs are thermally stable.

Part 2 of thesis

Results & Discussion

❖ Tensile Test





Tensile strength of gelatin films after the addition of starch

After adding PG starch tensile strength increased and addition AA starch increased the elongation at break

Results & Discussion

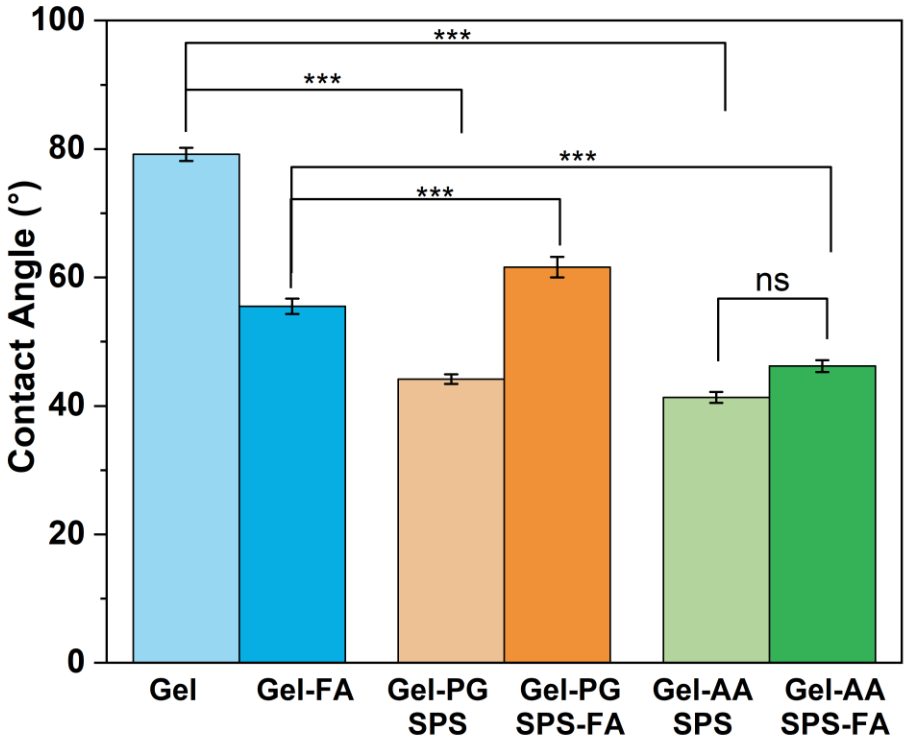
❖ Contact Angle

Hydrophilicity of ODF is measured

All the formulations were hydrophilic as contact angles were less than 90°

Incorporation of starch into gelatin film reduced the contact angle significantly

OH group of starch bonds with water and reduce the contact angle



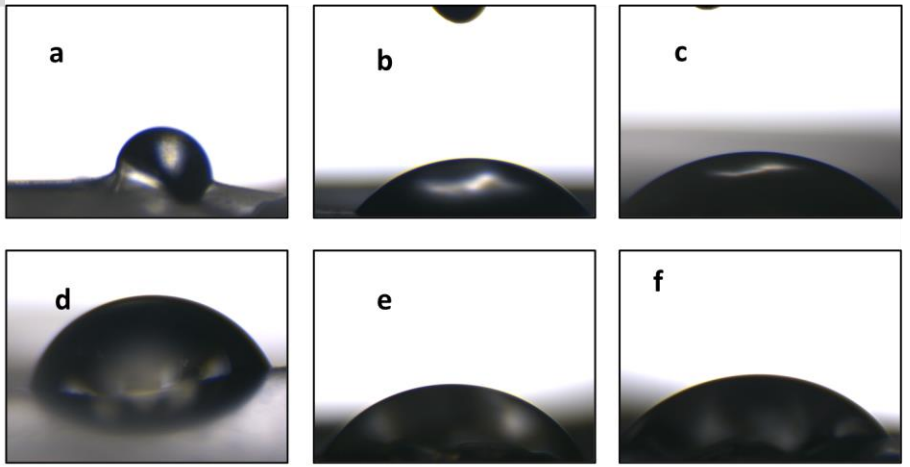
Results & Discussion

❖ Contact Angle

Among the starch-gelatin films, higher amylose content has positive correlation with film hydrophobicity

The films with high apparent amylose content have compact structure and less intermolecular space

Dense structure > Smaller intermolecular space> Larger contact angle > Film is not easily wetted by droplets



Part 2 of thesis

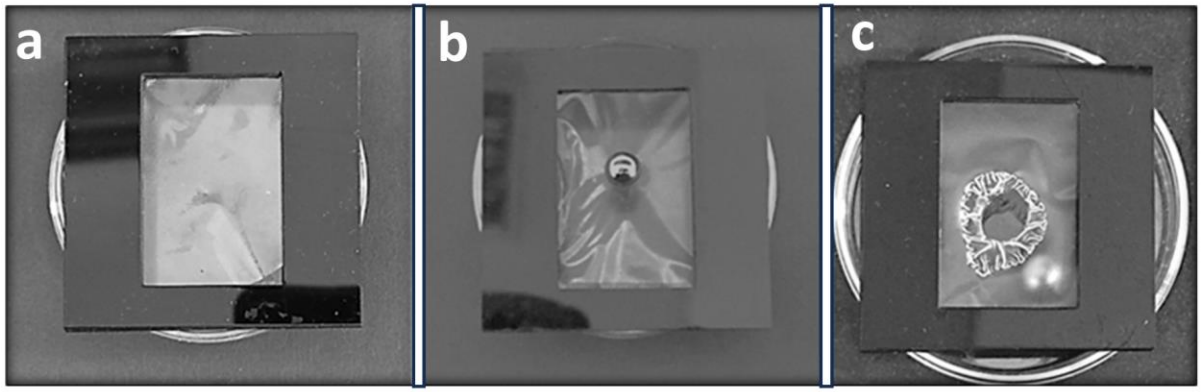
Results & Discussion

❖ Disintegration time

Disintegration time of ODF is one of the most crucial evaluating parameter.

Slide frame method is used with 200 mL simulated saliva as disintegration media

The 4g ball simulated the tongue force in the slide frame & ball method..



Several factors contribute to the disintegration time of ODFs, including particle size, polymer swelling, erosion, and hydrophilicity within the film matrix.

Part 2 of thesis

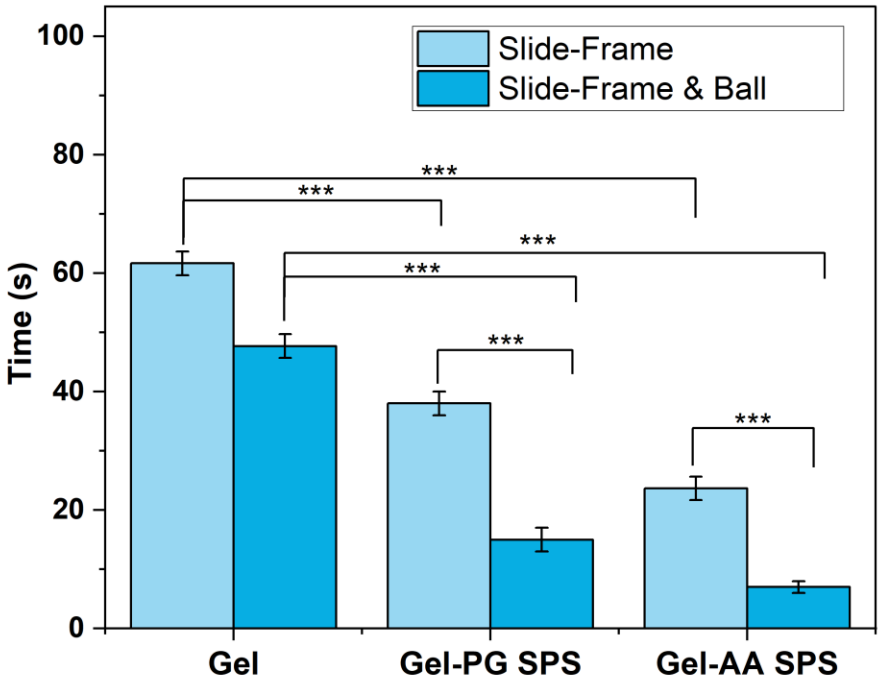
Results & Discussion

❖ Disintegration Time

Among all the ODFs, bovine gelatin film showed the highest disintegration time which is 61.67s ± 5.86 s

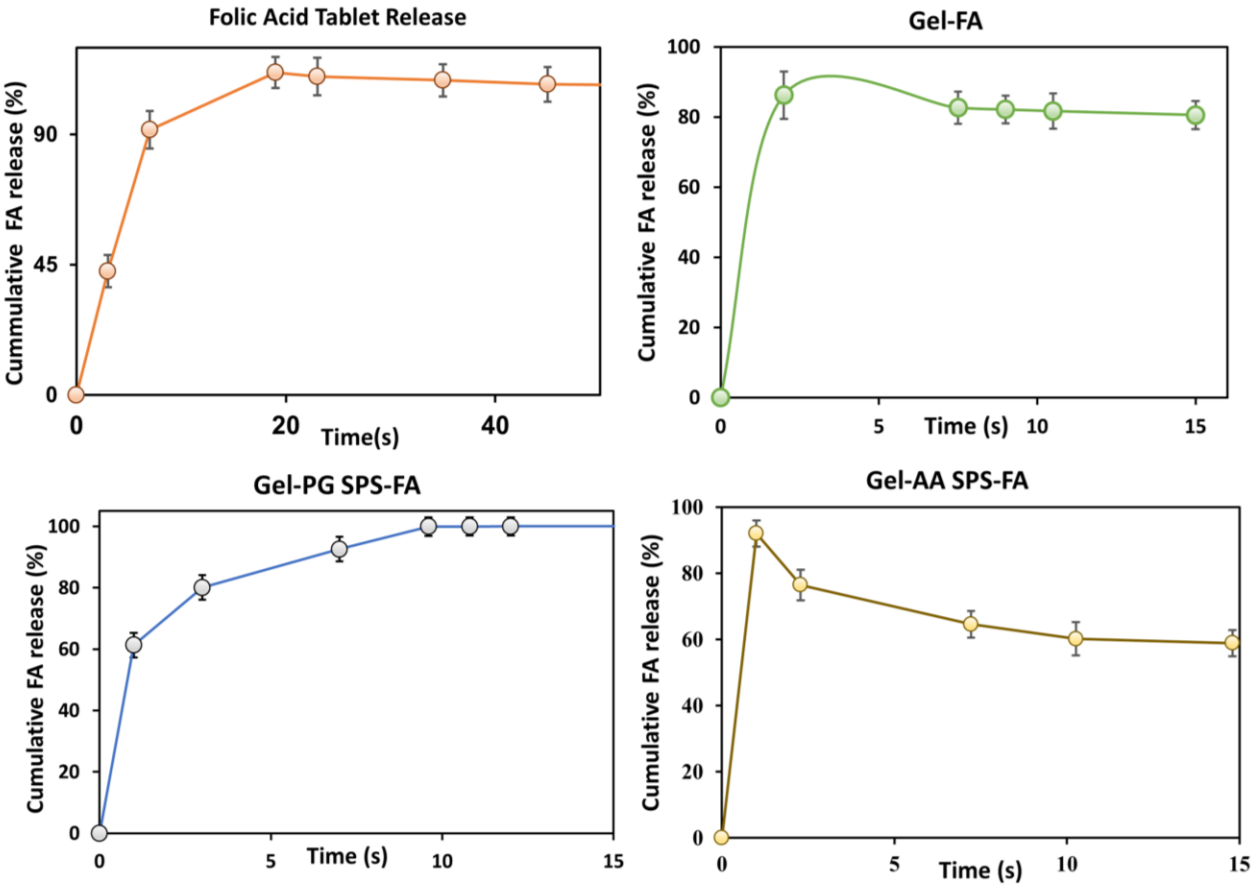
After incorporation of starch, gelatin/sweet potato starch films showed a faster swelling rate and erosion of the matrix

AA SPS showed the lowest particle size and least disintegration time



Results & Discussion

❖ Drug Dissolution



Results & Discussion

❖ Drug Dissolution

Formulation	First Order R^2	Gompertz R^2	First Order	Gompertz	
			k	α	β
Tablet	0.95	0.97	0.25	24.62	7.04
Gel-FA	0.14	0.94	0.86	0.14	-0.44
Gel-PG SPS-FA	0.94	0.95	0.76	0.52	2.81
Gel-AA SPS-FA	-0.032	0.95	2.21	0.19	-0.98

Oral mucosa can provide optimum condition of folic acid release without having any acidic pH

Fitted best with Gompertz model with a steep increase in the beginning and converges slowly to the asymptotic maximal dissolution

Among the ODF formulations, Gel-AA SPS-FA showed fastest drug release (92% release in 1 min)

Dissolution from ODFs is controlled by dissolution and erosion of film matrix

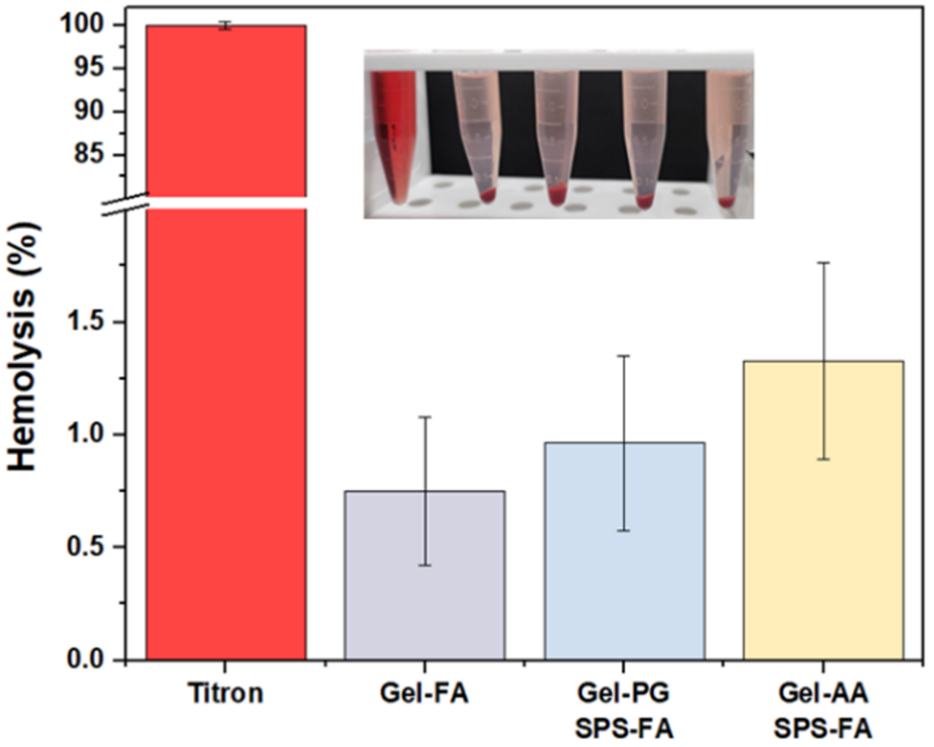
Results & Discussion

❖ Hemolytic Assay

A material is considered safe when the hemolytic efficiency is less than 5%.

ODFs have a very poor hemolytic activity of about 0.2%-1.6%.

ODFs are biocompatible for systemic circulation.



Summary of Part 2 of thesis

- A starch/gelatin composite oral disintegrating film was developed
- Depending on the starch modification, the gelatin/starch-based film showed optimized mechanical and functional properties
- Hydrophilicity of the ODFs were increased which showed reduced disintegration time and contact angle
- Mechanical properties of the ODFs were increased significantly as starch incorporation increased the tensile strength of the ODFs.

Conclusion & Recommendations

Scopes of the future work

- A burst folic acid release trend was shown. If a conjugate can be formed with the therapeutic, the release can be controlled and applications area can be more diverse
- Stability test of the developed ODFs should be done
- Hydrophilicity of the ODFs were increased which showed reduced disintegration time and contact angle

List of publications

1. Additive Manufacturing : A Bespoke Solution to Personalized Medicine, accepted in Rapid Prototyping Journal, Accepted at 16 May 2024

2. Role of starch modification on the performance and characterization of gelatin-starch based orally disintegrating film, Submitted

THANK YOU